$$\begin{bmatrix} 149 & 105 & 104 \\ 37 & 28 & 27 \\ 22 & 161 & 207 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$Small_x = (-1)(149) + (0)(105) + (1)(104) + (37)(-2) + 0 + 27\times 2$$
$$+ 22 \times -1 + 0 + 207$$

$$\cancel{= -163 + 551} \qquad = -149 + 104 - 74 + 54 - 22 + 207$$
$$\cancel{= 88} \qquad = 120$$

Sobel x = 120

$$Small_y = \begin{bmatrix} 149 & 105 & 104 \\ 37 & 28 & 27 \\ 22 & 161 & 207 \end{bmatrix} \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$$= -149 + (-2)(105) + (-104) + 22 + 161\times 2 + 207$$

$$= 88$$

$$\text{Sobel-combined} = \sqrt{Small_x^2 + Small_y^2}$$

$$\approx \underline{\underline{149}}$$